Theoretical Computer Science Exam, June 18th - July 2nd, 2018

The exam consists of **4 exercises**. Available time: 2 hr 00 min.; you may write your answers in Italian, if you wish.

All exercises must be addressed for the written exam to be considered sufficient.

F.NAME …………………………….. L.NAME …………………………………….. MATRICOLA …………………

I have passed the midterm exam and I intend to skip the first part (Ex.1-2) (select one): **YES NO**

Exercise 1 (7.5 pts)

Design minimal power automata accepting the languages defined as follows (as usual it is intended that, for every symbol k, the function $\#k(x)$ denotes the number of occurrences of k in the string x):

1) $L_1 = \{ x \in \{a, b, c\}^* \mid x$ contains no more than one pair of consecutive a's $\}$ For instance, $cbcaab \in L_1$, $aba \in L_1$, $cbaabcaab \notin L_1$, $cbaaab \notin L_1$.
2) $L_2 = \{ x \in \{a, b, c, d\}^* \mid \#a(x) + \#b(x) = \#c(x) + \#d(x) \}$ NB: solutions with few states and alphabet symbols are preferred.

Exercise 2 (7.5 pts)

Specify, by means of suitable pre- and post-conditions, a subprogram **inflate**(a, n, b, m) where:

1. the parameter a is an input array of positive integers having $n > 1$ elements, sorted in either increasing or decreasing order, and
2. b is an output array parameter, of size m, that also stores positive integers. (Indices in the arrays a and b start from 0)
3. the number of elements of array b is a multiple of n, i.e., it is equal to $k \cdot n$ for some $k>1$, and b stores the same elements as a, in the same order, each one multiplied by k and repeated k times; for instance, if $a = [5\ 7\ 9]$ then a possible value for b is [15 15 15 21 21 21 27 27 27] (in this case $k = 3$)

As a variation of the specification above, consider one in which the first two points are unchanged, and the third point is substituted by the following ones:

3. if a is sorted in increasing order then the first k elements of b are: $a[0] \cdot k$ followed by the next immediately successive k-1 integer values, in increasing order; the successive k elements of b are $a[1] \cdot k$ followed by the next immediately successive k-1 integer values, in increasing order, etc.; for instance, if $a = [5\ 7\ 9]$ then a possible value for b is [15 16 17 21 22 23 27 28 29] (in this case $k = 3$)
4. similarly, if a is sorted in decreasing order then the first k elements of b are: $a[0] \cdot k$ followed by the next immediately preceding k-1 integer values, in decreasing order; the successive k elements of b are $a[1] \cdot k$ followed by the next immediately preceding k-1 integer values, in decreasing order, etc. for instance, if $a = [9\ 7\ 5]$ then a possible value for b is [36 35 34 33 28 27 26 25 20 19 18 17] (in this case $k = 4$)

Exercise 3 (7.5 pts)

Consider a semidecidable, non-empty set S (so that it admits a generating function $g_S(x)$), and a decidable set D (so that its characteristic predicate $c_D(x)$ is computable), D being strictly included in S. Prove that the difference set $S - D$ is semidecidable. To this end, please define function $g_{S-D}(x)$, the generating function of set $S - D$.

How does the answer change if set D is not necessarily included in S?

How does the answer change if set S may be empty?

(Remember that the difference set of two sets A and B is defined as $A - B = \{ x \mid x \in A \wedge x \notin B \}$)

Exercise 4 (7.5 pts)

Let us recall the definition of a ***finer*** relation: a relation R' is said to be finer than another relation R iff

$$\forall x, y \, ((x \, R' \, y) \rightarrow (x \, R \, y)).$$

Then, consider two complexity functions $f(n)$ and $g(n)$. Function $g(n)$ is said to be an asymptotic upperbound for function $f(n)$, written in symbols as $f(n) \in \mathcal{O}(g(n))$, or equivalently $(f \, \mathcal{O} \, g)$, iff there exist a positive integer k and a positive constant c such that

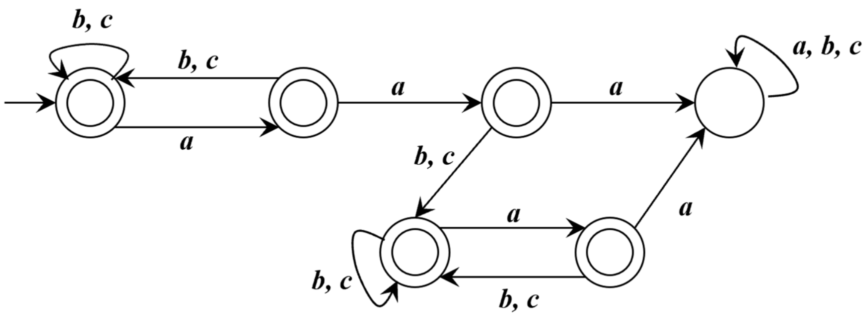
$$\forall n \ (n \geq k \rightarrow f(n) \leq c \cdot g(n))$$

1. Discuss, providing a suitable justification, possibly supported by simple examples, the following questions
   a. is relation Θ finer than $\mathcal{O}$?
   b. is relation $\mathcal{O}$ finer than Θ?
2. Relation $\mathcal{O}$ does not enjoy all the algebraic properties of relation Θ that are useful in the analysis of (time and space) complexity: specifically, which property of the Θ relation is not enjoyed by relation $\mathcal{O}$?

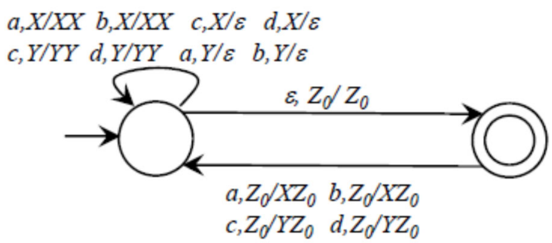
Solutions

Exercise 1

1. The language is regular, because during any computation it must "remember" a limited amount of information about the string read up to the current configuration.



2. A deterministic pushdown automaton suffices. In the following solution, a symbol X (respectively, Y) in the stack means that an eccess of symbols a and b (resp., c and d) have been read up to the current configuration. Notice that the automaton, is deterministic, despite the presence of a spontaneous transition.



Exercise 2

For the main case

pre: $n > 1 \land (\forall i \ (0 \leq i < n \rightarrow (a[i] < a[i+1])) \lor \forall i \ (0 \leq i < n \rightarrow (a[i] > a[i+1])))$

post: $\exists k \ (k > 1 \land m = k \cdot n \land \forall i \ \forall j \ (0 \leq i < n \land 0 \leq j < k \rightarrow b[k \cdot i + j] = k \cdot a[i] \ ))$

For the variation

post: $\exists k \ (k > 1 \land m = k \cdot n \land \forall i \ \forall j \ (0 \leq i < n \land 0 \leq j < k \rightarrow$

$$(a[0]<a[1] \rightarrow b[k \cdot i + j] = k \cdot a[i] + j \ )$$
$$(a[0]>a[1] \rightarrow b[k \cdot i + j] = k \cdot a[i] - j \ ) \ ) \ )$$

Exercise 3

Since S is non-empty and D is strictly included in S, then there exists an element K such that $K \in S\text{-}D$. Then the generating function of S - D is defined as follows

$$g_{S\text{-}D}(x) = \text{if } \ c_D(g_S(x)) = 0 \text{ then } g_S(x) \ \text{ else } K$$

(notice that $c_D(g_S(x)) = 0$ means that $g_S(x) \notin D$)

If set D is not strictly included in S the answer does not change. Two cases arise.
1. $S \subseteq D$: then $S - D = \emptyset$ and S is semidecidable by definition.
2. $(S \cap D) \subset S$ and $(S \cap D) \neq S$; hence there exists an element $K \in S - D$ and the argument follows the same lines as above.

If set S is empty, then so is $S - D$, which is therefore semidecidable by definition.

Exercise 4

1. Discuss, providing a suitable justification, possibly supported by simple examples, the following questions
   a. is relation Θ finer than $\mathcal{O}$?
   b. is relation $\mathcal{O}$ finer than Θ?

If $f\ \Theta\ g$ then $\lim_{n\to\infty} \frac{f(n)}{g(n)} = c$ for some c such that $0 < c < \infty$, therefore[1] $\exists k$, $\exists c'$ such that $\forall n\ (n \geq k \rightarrow f(n) \leq c' \cdot g(n))$, hence $(f\ \mathcal{O}\ g)$ and Θ is finer than $\mathcal{O}$. Consider for instance $f(n)=3n^2+n$ and $g(n) = 2n^2-n$: both the following hold $(f\ \Theta\ g)$ and $(f\ \mathcal{O}\ g)$.

On the other hand, however, $(f\ \mathcal{O}\ g)$ does no in general imply $(f\ \Theta\ g)$; as a counterexample, consider $f(n) = n$ and $g(n) = n^2$: then certainly $(f\ \mathcal{O}\ g)$ but $(f\ \Theta\ g)$ does not hold.

2. Relation $\mathcal{O}$ does not enjoy all the algebraic properties of relation Θ that are useful in the analysis of (time and space) complexity: specifically, which fundamental property of the Θ relation is not enjoyed by relation $\mathcal{O}$?

Relation $\mathcal{O}$ is ***not*** symmetric (i.e., $(f\ \mathcal{O}\ g)$ does not imply $(g\ \mathcal{O}\ f)$): as a counterexample consider again $f(n) = n$ and $g(n) = n^2$: then it holds $(f\ \mathcal{O}\ g)$ but $(g\ \mathcal{O}\ f)$ does not hold.

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[1] $\left|\frac{f(n)}{g(n)} - c\right| < \varepsilon$ implies $\frac{f(n)}{g(n)} < c + \varepsilon$ hence $f(n) < (c+\varepsilon)g(n)$